



Auto-ignition characteristics of a near-term light surrogate fuel for marine diesel: An experimental and modeling study

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ABSTRACT

Autoignition characteristics of a four-component surrogate fuel(M2) were investigated in a heated rapid compression machine over a wide range of conditions. The near-term light surrogate M2 developed for marine diesel fuel was formulated in our previous study containing 17.3% n-hexadecane, 5.93% 2,2,4,4,6,8,8-heptamethylnonane, 30% decalin and 46.77% 1-methylnaphthalene by mole fraction. Ignition delay times (IDTs) of gas-phase M2/N₂/O₂ mixture were measured at pressures of 7, 10, 15 and 20 bar over the temperature range from 675 K to 914 K for the equivalence ratios (ER) of 0.5, 1 and 1.5. NTC behaviours within the temperature of 730–850 K and two-stage ignition phenomena were observed. The effects of operating conditions such as pressure, equivalence ratio and oxygen content were systematically studied, and correlations of total IDTs and first-stage IDTs were performed to further quantitatively reveal the IDTs dependence on the above parameters. Reasonable modifications were made to a literature mechanism to improve its predicting quality. The simulation results using the modified model could satisfactorily reproduce the IDTs under the test conditions, and the model is also able to capture the ignition delay dependence on pressure and fuel/oxygen content. A brute force sensitivity analysis at the low temperature (680 K) and the NTC temperature (780 K) was carried out to identify the key reactions that govern the ignition event of surrogate M2.

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1. Introduction

Cargo ships equipped with large two-stroke marine diesel engines account for the majority of worldwide maritime transport [1]. Marine diesel engines are powered by heavy fuel oils(HFOs) and the combustion characteristics of fuels in combustion chambers are the decisive factor for the performance of engines [2]. Thus, understanding and controlling the combustion process of fuels from the fundamental side is essential for future engine developments. Nowadays, Computational Fluid Dynamics(CFD) coupled with precise fuel kinetics provides a short-cycle, low-cost and high-accuracy method to understand the interaction of chemical reactions and flows, as well as providing guidance for the design and optimization of future engines [3]. CFD simulations are especially important for the improvement of large marine engines due to the challenges of performing experimental tests. In the simulation process, surrogate fuels with fewer components are applied to mimic the combustion behaviours of real fuels in practical engines, replacing real-fuels to reduce the computational complexity.

A successful surrogate fuel should match the key physical-chemical properties of the target fuel according to the intended application. Several reviews have summarized the previous efforts that have been paid to develop surrogate fuels for gasoline [4], diesel [5] and jet fuels [6]. It is worth noticing that the ultimate goal of surrogate formulation is to develop high-fidelity chemical-kinetic models which are capable of being applied to engine simulations.

Unlike the automotive fuel and jet fuel, the marine diesel oil is a kind of black, high-viscosity, high impurity-content and low-cost liquid fuel [7]. This kind of fuel consists of many extremely large hydrocarbon molecules whose carbon numbers mainly distribute in the range of 18–28 [8]. So representative compounds which belong to this carbon range should be selected when constructing comprehensive surrogate fuel for HFOs, in order to match the key physical and chemical properties of the target fuel. However, the chemical kinetic models of the compounds falling within this range are less developed till now, and the kinetic mechanism, thermodynamic data and transportation data of most of these molecules do not even exist currently. That is also the major challenge for building surrogate fuels for HFOs. Farrell et al.[3], put forward the idea of long-term surrogate formulation and near-term surrogate formulation for real fuels. The long-term goal of surrogate fuels is

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to represent full blend fuel chemical kinetics, consequently which would require more complex components than can be handled in current computational codes. Therefore, the near-term research should reflect a long-term desire to simultaneously describe both physical and chemical kinetic parameters of real fuels, based on the current development of chemical-kinetic level and available simulation sources. Thus, the development of surrogate fuels for heavy fuels should meet the current demand for engine simulation and meanwhile balance the matching quality with the up-to-date kinetic development.

So far, few studies have focused on the development of surrogate fuels for heavy marine diesel fuels. In general, some single-component or simple multi-component surrogate fuels are usually used in the CFD simulation of large two-stroke marine diesel engines. For instance, *n*-heptane [9,10] and *n*-tetradecane [11,12] are often used as surrogates for marine diesel fuels. Sun et al. [13] developed a reduced *n*-tetradecane-toluene mechanism as a surrogate fuel for marine diesel engines. The mechanism included 81 species and 317 reactions and it was validated against data collected from experiments in shock tubes, flow reactors, jet-stirred reactors and marine engines. However, obviously, the above mentioned surrogate fuels are the oversimplification of real HFOs. As they can neither match the real fuel properties nor cover the representative hydrocarbon class in this kind of actual fuels. More recently, Sun et al. [14] proposed a new surrogate fuel for HFOs consisting of eight-components: $C_{14}H_{16}$, $C_{16}H_{34}$, $C_{20}H_{42}$, $C_{21}H_{44}$, C_5H_{10} , C_7H_8 , $C_{10}H_8$ and $C_{14}H_{10}$. The inclusion of wide ranges of compounds in the new surrogate improved the matching quality on physical-chemical properties. However, the researchers only validated the simulation results against the experimental data of the single component from Shock Tube and did not perform validating experiments using the proposed surrogate mixture. Jameel et al. [15] developed surrogate fuels for two kinds of marine diesel oils using the method of Nuclear Magnetic Resonance (NMR). Firstly, NMR was applied to analyzing the representative molecular structure contained in the target fuel, and then they construed a new molecule as the surrogate fuel for HFOs according to the analytical results. Though the new molecule represents the structure characteristics of HFOs, it can be unrealistic to develop its kinetic models due to its too complicated structure.

Wu et al. [16] systematically analyzed the composition and key properties of a kind of heavy marine diesel (RMG180) and two surrogate fuels were generated by emulating the representative properties and compositional characteristics of the target fuel. The heavy long-term surrogate fuel M1 was generated based on fitting volatility (distillation curve), density and ignition quality, and another light near-term surrogate fuel M2 was obtained by fitting ignition quality, density and the H/C ratio. In order to match the distillation property of RMG180, M1 contains some macromolecular components with high boiling points, however, the kinetic models of which are still unavailable. Thus, four compounds (17.3% *n*-hexadecane, 5.93% 2,2,4,4,6,8,8-heptamethylnonane, 30% decalin and 46.77% 1-methylnaphthalene by mole fraction) whose kinetic models are available at present were selected for M2, so as to meet the demand of developing corresponding kinetic models in the near future. In this study, further experimental and simulation investigations are carried out to study the auto-ignition characteristics and the kinetic mechanism of surrogate M2.

The current study aims to investigate the ignition delay times (IDTs) of surrogate M2 and optimize its kinetic model. The four-component surrogate M2 is tested on a heated Rapid Compression Machine (RCM) under low-to-intermediate temperature (*T*) range, and its ignition characteristics were systematically studied. The effects of pressure, equivalence ratio (ER), and oxygen mole fraction on IDTs were experimentally investigated and quantitatively analyzed using correlation methods. Then, a semi-detailed mechanism

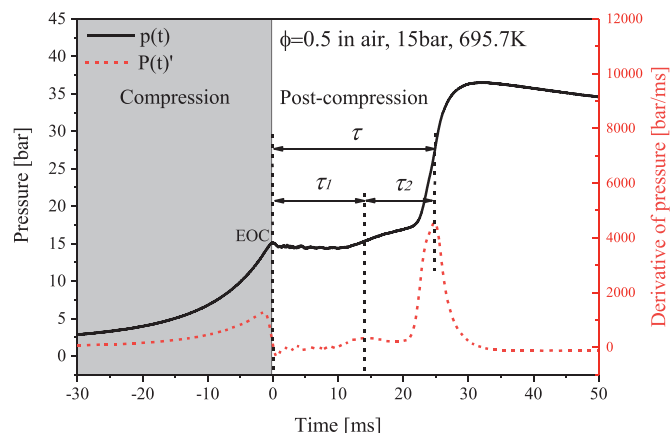


Fig. 1. Definition of ignition delay time (IDT) in the rapid compression machine.

is modified to improve the performance in predicting the auto-ignition characteristics. The experimental results are compared to the simulation results of the modified mechanism. Finally, sensitivity analysis is performed to identify the critical reactions that greatly affect the overall reactivity.

2. Methods

2.1. RCM measurement

The IDTs of surrogate M2 in the low-to-intermediate temperature range were measured in a heated RCM at Shanghai Jiao Tong University. The details of the experimental setup were introduced in our previous studies [17–19], so here we just briefly describe it. The fuel tank, the combustion chamber, the pipes and all the valves are wrapped with preheating jackets that are heated and maintained at a maximum temperature up to 180 °C in the whole testing process, to make sure the reactive mixture is fully mixed and vaporized. After the mixture entered into the chamber, the piston is driven into the cylindrical chamber by compressed air within 30–45 ms, and then a virtually constant-volume adiabatic process happens. The pressure trace in the RCM is measured by a pressure transducer (Kistler 6125B) and amplified by a charge amplifier (Kistler 5015). A LabVIEW program was designed and used for experiment initiation and data acquisition.

In the IDT measurement, the compressed pressure (P_c) and the compressed temperature (T_c) are two basic parameters. The former is directly obtained from the experimental pressure profile, and the latter is calculated according to Eq. (1) based on the ‘adiabatic core assumption’ [20]. In the equation, T_0 and P_0 represent the initial temperature and initial pressure respectively. γ refers to the specific heat ratio ($\gamma = C_p / C_v$) of the fuel/oxidant mixture. The compressed pressure (P_c) is adjusted by changing the compression ratio and the initial pressure.

$$\int_{T_0}^{T_c} \frac{\gamma}{\gamma - 1} \frac{dT}{T} = \ln \left(\frac{P_c}{P_0} \right) \quad (1)$$

Fig. 1 illustrates the definition of measured IDTs in RCM. The black solid line is a typical two-stage ignition pressure trace and the red dash line shows its first derivative. The total IDTs (τ) is defined as the time between the end of the compression and the local maximum of the first derivative of pressure, and its value is equal to the sum of the first-stage IDT (τ_1) and the second-stage IDT (τ_2). τ_1 is the time interval between the end of compression and the first local maximum of the pressure derivative. τ_2 is the time interval between the first local maximum and the second local maximum of the pressure derivative. Each experimental point

Table 1
Formulation of surrogate M2.

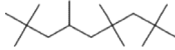
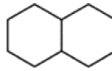
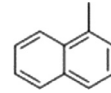
Compounds	Surrogate M2			
	n-hexadecane	2,2,4,4,6,8,8-heptamethylnonane	Decalin	1-methylnaphthalene
Formula	C16H34	C16H34	C10H18	C11H10
Structure				
Mol%	17.3	5.93	30	46.77
Wt%	24.39	8.36	25.83	41.42

Table 2
Comparison of Properties of target fuel(RMG180) and the surrogate(M2).

Properties	Marine diesel, RMG180	Four-component surrogate, M2
CN	40.96	41.12
H/C ratio	1.62	1.51
Density at 20 °C[Kg/m ³]	940.0	883.2

is repeated at least three times and the ignition data which is closest to the average value is reported. The uncertainty of RCM measurement is mainly caused by the reactant preparation, the pressure measurement and the determination of temperature, and finally, the overall uncertainty of IDTs is estimated to be $\pm 10\%$ [21].

2.2. Reactant preparation and test conditions

The tested fuel in this study is the marine diesel surrogate fuel M2 which consists of four-components including n-hexadecane, 2,2,4,4,6,8,8-heptamethylnonane (isocetane), decalin and 1-methylnaphthalene, and Table 1 lists its composition and the molecular structures of the components. The detailed formulation method for M2 was demonstrated in the previous study [16]. Herein, Table 2 shows the comparison of the target properties (CN, H/C ration and density) between RMG180 and M2. The four compounds could cover the main hydrocarbon classes (n-alkanes, isoalkanes, cycloalkanes and aromatics). Other than the compositional characteristics, both the chemical and physical properties are taken into consideration when generating the surrogates. In order to develop a near-term surrogate for the heavy fuel oil based on the current level of kinetic development, the kinetic models of the selected compounds are all currently available and belong to the large-molecule group in their corresponding hydrocarbon classes.

Preheating is necessary for RCM experiment to provide gas-phase reactant. Here the mixture is heated and maintained at 150 °C in the whole test process to ensure that the partial pressure of a single component is much less than its saturation pressure. The partial pressure data of components for all the test conditions are provided at Table S1 in the supplemental material. Besides, it is also worth noting that there are huge challenges to conduct IDT measurements for heavy marine diesel mainly due to its extremely high boiling points. That is also one of the reasons why few experimental data for large hydrocarbons ($\geq C_{20}$) exist. During this test, the partial pressure of fuel M2 after its injection showed that evaporation rates of all the conditions were more than 99%. The four components are provided with high purities and their procurement information are presented at Table S2 in the supplemental material. M2 is blended in advance by mass fraction and then injected into the fuel tank. Nitrogen and oxygen are added to the tank through a pressure gage, and the purity of them is more than

99.999%. Finally, the mixture is left for more than 2 h for fully mixing.

Table 3 presents the experimental conditions and the composition of the reactants. Ignition delay times for surrogate M2 were measured at a pressure of 7, 10, 15 and 20 bar over a range of temperature from 675 to 914 K, and for equivalence ratio from 0.5 to 1.5. With the proper arrangement of the proportion of the reactant mixture, the IDT dependence on temperature, pressure, equivalence ratio, and oxygen content is systematically investigated. All of the reported RCM IDTs are provided in the supplementary material.

2.3. Numerical methods

In the current study, the CHEMKIN-PRO software was used to perform kinetic simulations to predict the IDTs under the conditions in Table 3. There are two kinds of modeling methods involved here, called constant volume simulation (CONV) and variable volume simulation (VARV). CONV refers to the case that the reactant in a fixed-volume and adiabatic reactor react at an initial temperature and pressure. The energy equations are solved to obtain the temperature, pressure and species duration and other parameters within the reaction system, and does not take the compression process and heat loss into account. In contrast, VARV was performed to possibly restore the real process in RCM [22,23]. The compression history and post-compression heat loss were considered by converting the measured pressure duration into the volume variation based on the adiabatic core assumption and the ideal gas law. The conversion from measured pressure to volume was calculated according to Eq. (2) as shown in the following:

$$V_t = V_0 * \left(\frac{P_t}{P_0} \right)^{\frac{1}{\gamma}} \quad (2)$$

V_t , V_0 , P_t and P_0 refer to the effective volume, the initial volume, the measured pressure and initial pressure, respectively. A non-reactive test is performed for each temperature point so as to record the pressure traces P_t that is used to calculate the volume profile V_t . Moreover, it should be indicated that performing VARV simulation is far more time-consuming than doing CONV simulation.

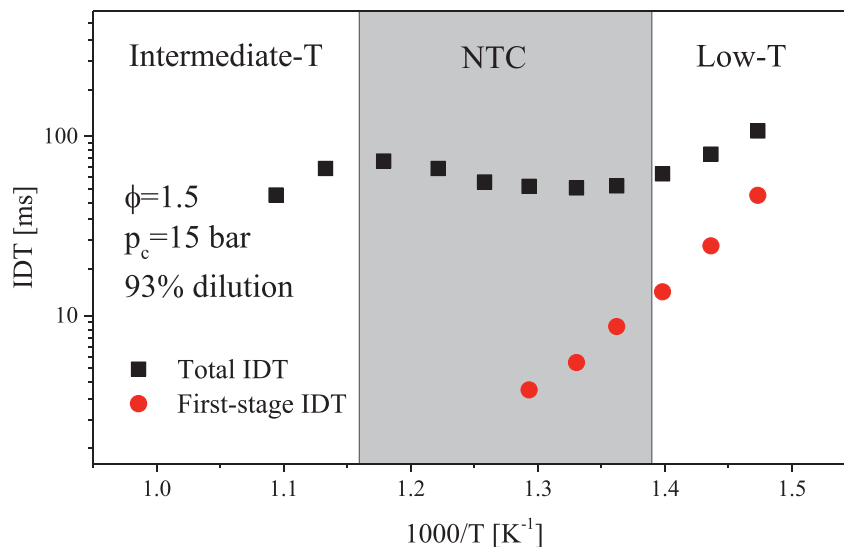
Besides, the sensitivity analysis was performed to identify the controlling reactions that affect much on the total ignition delay times. The sensitivity coefficient S_i is defined in Eq. (3), shown in the following:

$$S_i = \frac{\tau(2k_i) - \tau(k_i)}{\tau(k_i)} \quad (3)$$

Among the equation, k_i denotes the reaction constant of reaction i . and $\tau(k_i)$ is the simulated total IDT of reaction i . $\tau(2k_i)$ refers to its total IDT when the rate constant of reaction i is multiplied by a factor of 2. According to the definition, the greater the absolute value of the sensitivity coefficient, the greater the effect

Table 3
Reactants and test conditions.

NO.	Φ	X_{fuel} [mol%]	X_{O_2} [mol%]	X_{N_2} [mol%]	Dilution ratio $X_{\text{N}_2}/(X_{\text{N}_2}+X_{\text{O}_2})$ [%]	Pressure [bar]
01	0.5	0.64	20.87	78.49	79	10/15
02	1	1.27	20.74	77.99	79	7/10/15
03	1.5	1.89	20.61	77.50	79	7/10
04	1	0.64	10.44	88.93	89.5	15/20
05	1.5	0.64	6.96	92.4	93	15/20

**Fig. 2.** IDT of surrogate M2 over low-to-intermediate temperature range ($\Phi=1.5$, 93% dilution, $P_c=15$ bar).

of the reaction on the total IDTs, and the positive or negative of S_i respectively indicate that reaction i has inhibited or promoted effects on the reactive system.

3. Results and discussion

3.1. Auto-ignition characteristics

Fig. 1 shows the pressure curve (black solid line) and its first-order derivative (red dotted line) of the M2/air mixture at an equivalent ratio of 0.5, a pressure of 15 bar, and a temperature of 695.7 K. The figure shows the typical two-stage ignition characteristics of M2, indicating that M2 possesses low-temperature reaction pathways. According to the previous studies [17,24–26], two-stage ignition phenomena was found in experimental investigations of n-hexadecane, isocetane and decalin, however, no two-stage ignition was found for methylnaphthalene. Even though the content of methylnaphthalene in M2 is the highest (41.42% wt) among the components, M2 still shows obvious two-stage ignition similar to the former three components, which indicates the dominant role of paraffins and naphthenes in low-T reactions.

Fig. 2 shows the total IDTs and the first-stage IDTs of the M2 at 15 bar for an equivalent ratio of 1.5. As shown in the figure, the surrogate fuel displays an evident negative temperature effect (NTC) phenomenon. In the temperature range from ~730 K to ~850 K (the shadowed part), the total IDTs increase with increasing temperature, while IDTs in the Low-T and Intermediate-T range (the white background part) decrease with increasing temperature. Both total IDTs and first-stage IDTs exhibit a linear relationship with $1000/T$ in the logarithmic coordinate system in the Low-T range, which can be also consistent with the performance of most hydrocarbon fuels [27–29]. In contrast, the slope of the first-stage IDTs is larger than that of the total IDTs, which shows that

first-stage IDTs is more sensitive to temperature according to the Arrhenius law of the elementary reaction rate.

To further analyze the NTC phenomenon and the two-stage ignition characteristic, Fig. 3 displays the pressure histories of all the temperature points in Fig. 2. Similarly, the pressure curves are classified into three parts according to the T range. On the whole, the peak pressure dropped with the rise of temperature, which is contributed by the more severe heat loss under higher temperature. At the temperature of 678.8 K, the pressure shows a slight first-stage increase after compression, and then hot ignition occurs and pressure suddenly rises, demonstrating the typical two-stage ignition phenomenon. As the temperature continues to increase, first-stage IDTs become shorter until 794.8 K in the NTC interval, and obvious two-stage ignition phenomenon disappears. In this process, the first-stage pressure increase becomes larger when the temperature rises from 678.8 K to 715.2 K, but then its value gradually decreases till the temperature reaches 794.8 K. The continuous shortening of first-stage IDTs is due to the continuous growth of the low-temperature reaction rate with the increase of temperature, which directly leads to the reduction of the total IDTs in Low-T region. In the NTC range, the decomposition of QOOH to alkenes and HO₂ gradually becomes competitive to the O₂-addition to QOOH. Thus, Low-T chain branching pathways would become less dominant and stable intermediates, such as alkene and HO₂ are accumulated, leading to a lower reactivity. So, the total IDTs in NTC interval increase as temperature goes up. However, when temperature enters into the intermediate temperature range, the low-temperature pathways gradually shut off and the high-temperature reactions become dominant. At the same time, the reactions accelerate as temperature rise, so the total IDTs decrease. Generally, low-temperature chain-branching reactions in fuel oxidation lead to the NTC behaviours and the two-stage ignition characteristics [30].

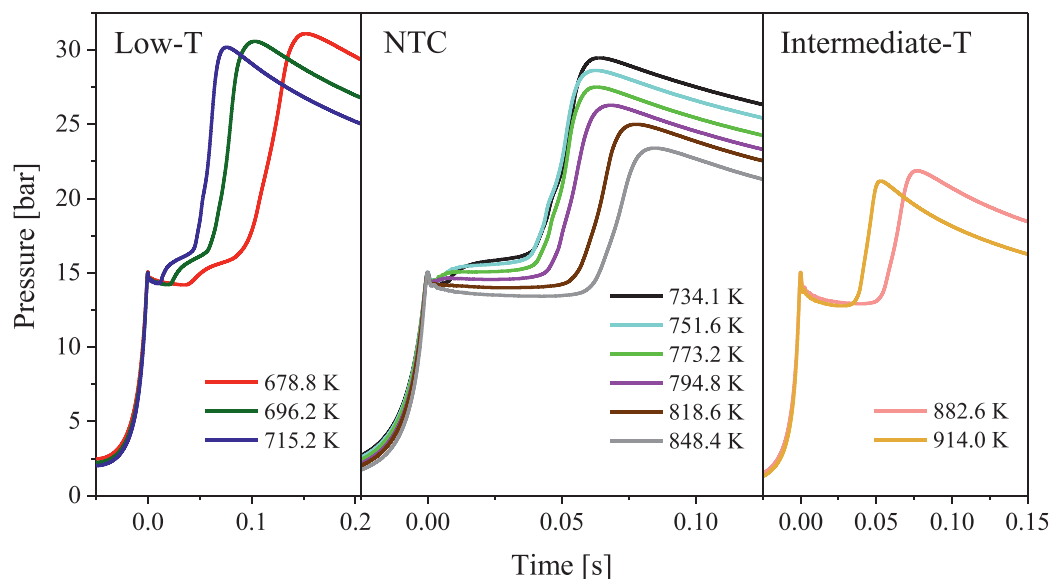


Fig. 3. Pressure traces of surrogate M2 under the condition of $\Phi=1$, 89.5% dilution and $P_c=10$ bar.

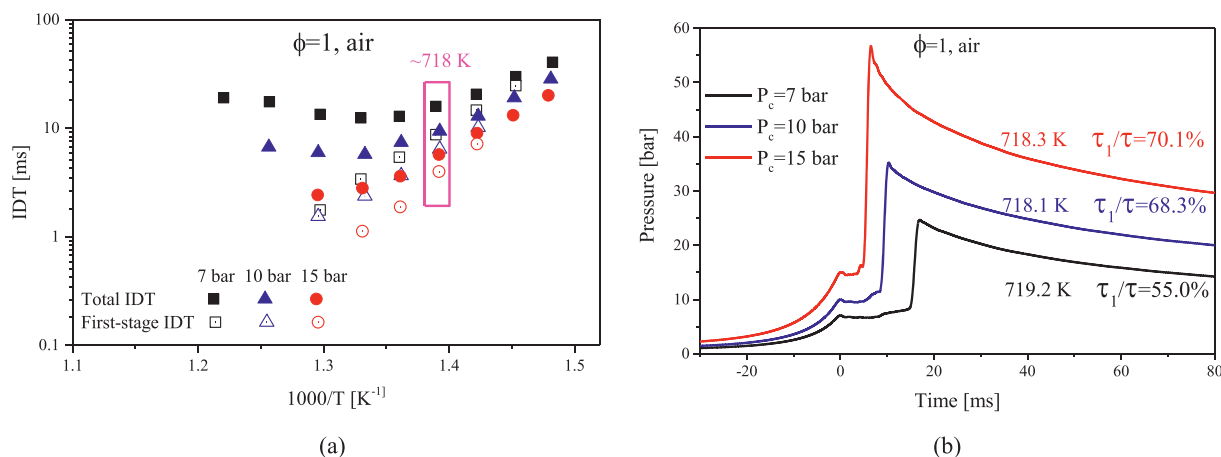


Fig. 4. Comparison of ignition characteristics of surrogate fuel M2 under different pressure: (a) comparison of ignition delay times, (b) comparison of measured pressure curves.

3.2. Effects of operating conditions on IDTs

3.2.1. Effect of pressure on IDTs

Fig. 4 presents the IDTs and the pressure histories of M2/air under different pressures. As shown in Fig. 4(a), both the total ignition delay times and the first-stage ignition delay times decrease with the increasing pressures. This is mainly due to that the higher pressure is achieved by the higher initial reactant concentration in the reacting system. The increased fuel/air mixture concentration contributes to improved reaction rates as well as more heat release, which ultimately leads to a decrease in IDTs. However, the effects of pressure on the IDTs in different temperature ranges are slightly different. It can be seen that the pressure shows little influence on the IDTs in the Low-T range. As the temperature increases and enters the NTC region, the IDTs at different pressures show large discrepancies and the IDTs dependence on pressure gets much stronger. NTC phenomenon becomes less pronounced at higher pressure and the starting temperature of NTC phenomenon shift to a higher temperature when pressure increase. Besides, the first-stage IDTs are less sensitive to pressure change compared to the total IDTs. The similar trend was also found in some previous studies for other hydrocarbons [31–34]. This trend is highly rel-

evant to the addition of O_2 to form RO_2 and the dissociation of RO_2 , therein the former is a crucial step in the low-T chain branching pathways and the latter would be rather important at high temperature. According to the Le Chatelier effects, higher pressure would promote the low-T reaction $R + O_2 = RO_2$, therefore, the low-T reactivity would continue to be important at a higher temperature, which could lead to a higher starting temperature of NTC behavior. At the same time, the dissociation of RO_2 would be inhibited and the H_2O_2 decomposition would be accelerated as pressure increases [31]. Thus, the NTC behaviors will become less pronounced.

Fig. 4(b) compares the pressure histories of one temperature point in Fig. 4(a) under the pressure of 7/10/15 bar at around 718 K. It can be seen that both the total IDTs and the first-stage IDTs decreased with increasing pressure. Besides, pressure traces of higher pressure present a greater first-stage pressure-climb and a much steeper pressure-rise in the hot ignition. This also results from the higher reactant concentration under higher pressure condition. In addition, the proportion of 1st IDTs in the total IDTs drops when pressure decreases. It means that the total IDTs grows faster than the 1st IDTs as pressure reduce, which indicates that the total IDTs is more sensitive to pressure change.

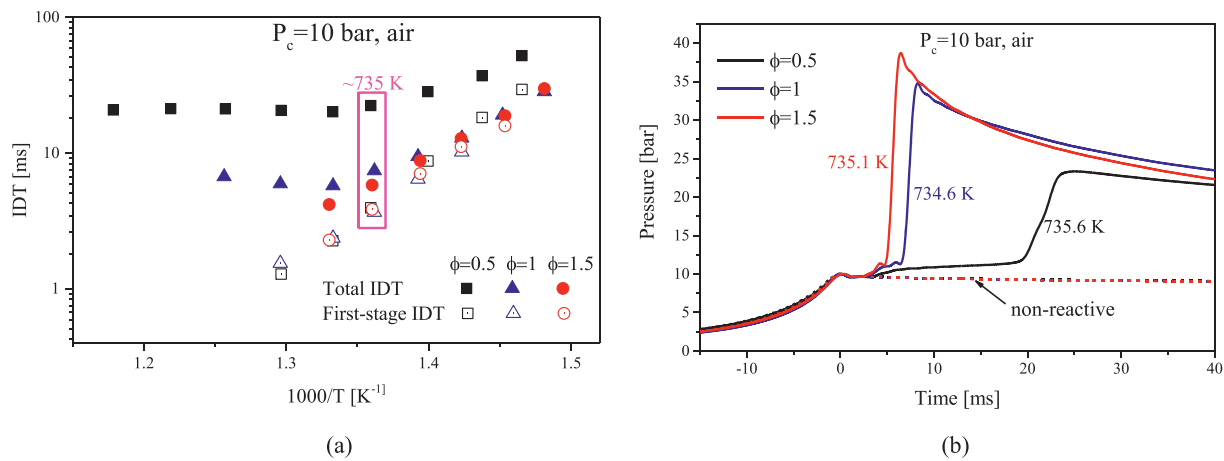


Fig. 5. Comparison of ignition characteristics of surrogate fuel M2 under different equivalence ratios: (a) comparison of ignition delay times, (b) comparison of measured pressure curves.

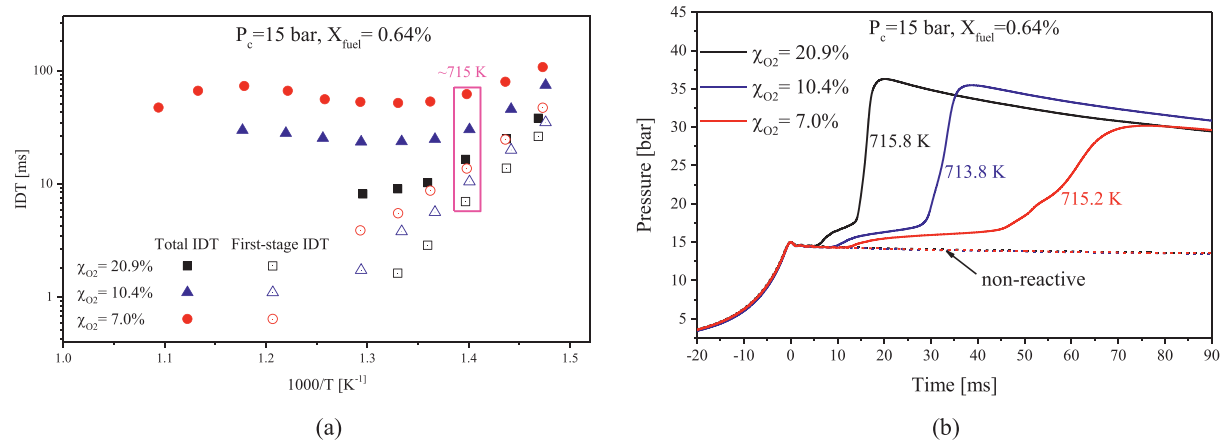


Fig. 6. Comparison of ignition characteristics of surrogate fuel M2 under different oxygen content: (a) comparison of ignition delay times, (b) comparison of measured pressure curves.

3.2.2. Effect of equivalence ratio on IDTs

Fig. 5 shows the comparison of ignition characteristics of M2/air mixture under the pressure of 10 bar at different equivalence ratios. As is illustrated in Table 3, different equivalence ratios were realized by changing the fuel content and maintaining the oxygen content at a constant value (about 20.7%). So herein the effect of equivalence ratio equals to the effect of fuel content. Generally, the increase in the equivalence ratio can lead to a decrease in the total IDTs. In the low-T region, the total IDTs of fuel-rich ($ER=1.5$) case and the stoichiometric ($ER=1$) case are almost the same and both cases are significantly lower than the fuel-lean ($ER=0.5$) case, which may be caused by the greater heat loss in the case of the lean mixture. Moreover, the equivalence ratio exhibits a stronger influence on the IDTs in NTC region than IDTs in the low-T region. The reason of this trend is that in the low-T range, the increased fuel concentration produces more fuel radicals that dominate the low-T chain branching pathways via O_2 -addition reactions at constant oxygen content. In the NTC range, besides the above low-T pathways, the reaction class ' $RH + HO_2 = R + H_2O_2$ ' will also be promoted, which could lead to the acceleration of the decomposition of H_2O_2 . Thus, the decline of IDTs in the NTC regime is more pronounced when fuel contents increase.

In Fig. 5(b), the pressure traces for the M2/air at the temperature of 735 K under the pressure of 10 bar for different equivalence ratios are displayed with their corresponding non-reactive pressure profiles. It can be seen that there is little difference between the

three pressure curves in the compression stroke due to the larger C_p of fuel compared to N_2/O_2 . Meanwhile, the non-reactive pressure traces in the post-compression regime showed no obvious difference between the three curves, which further manifest the reproducibility of the experimental setup. The peak pressure goes up with the increase of equivalence ratio, whereas the pressure rise of the first stage doesn't show distinct disparities. The higher equivalence ratio means higher fuel content, thus resulting in an increase of reaction rate and total heat release, which can directly enhance the pressure rise in the hot ignition. However, the similar pressure climb in the 1st stage ignition demonstrates that the equivalence ratio has little influence on the heat release in the 1st stage ignition.

3.2.2. Effect of oxygen content on IDTs

The IDTs dependence on oxygen concentration is shown in Fig. 6. Fig. 6(a) compares the total IDTs and 1st IDTs under the compressed pressure of 15 bar with different oxygen content from 7.0% to 20.9% when the fuel content is fixed. It can be observed that IDTs shorten as the oxygen content increases in the low-to-intermediate region. According to the studies [35,36], the low-temperature chain-branching pathways of fuels consists of the two O_2 addition steps, isomerization, and decomposition, producing reactive hydroxyl radicals. It is evident that increasing oxygen content could improve the reactivity of low-T reactions and lower the ignition delay times in the reacting system. Also, the effect of

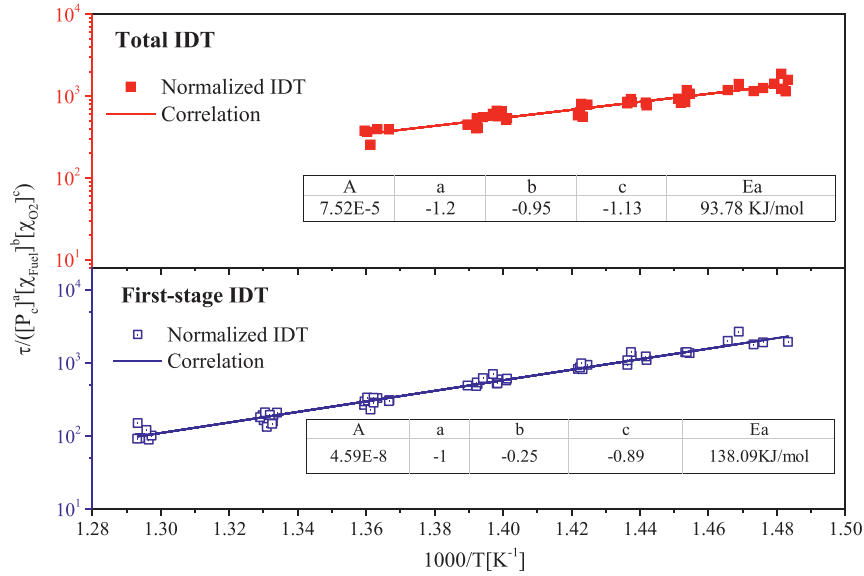


Fig. 7. Correlations of total IDTs in the low-T region and first-stage IDTs.

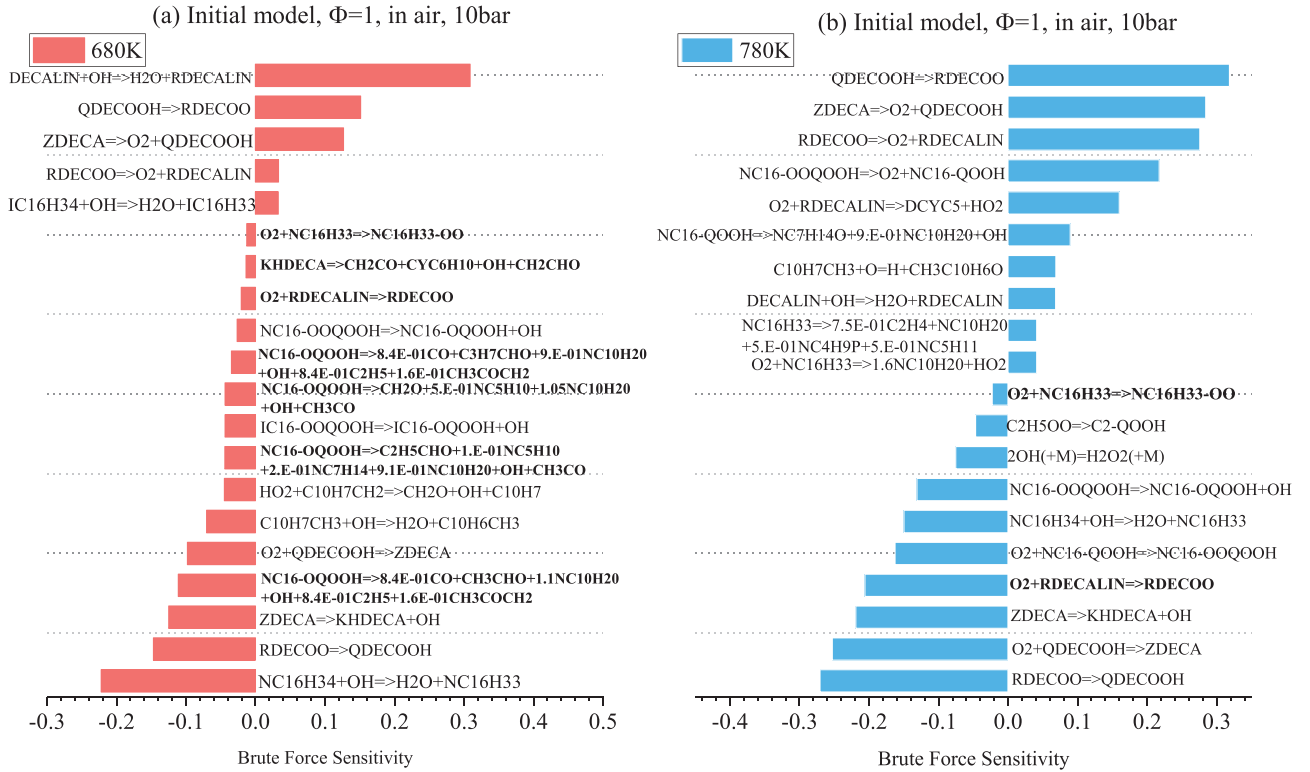


Fig. 8. Sensitivity analysis using the initial model: (a) 680 K, (b) 780 K.

oxygen content on IDTs gets stronger as the temperature moves to the intermediate-T range from low-T range.

Fig. 6(b) further illustrates the effect of oxygen content by showing the pressure traces at one of the temperature points (~715 K) in Fig. 6(a). It is shown that the pressure climbs faster and the peak pressure is higher when oxygen content goes up, which indicates that more oxygen can result in more heat release thus leading to quicker pressure rise and larger final pressure. However, the pressure rise in the 1st stage ignition is not affected much, which means that no obvious change of the heat release in the 1st stage ignition happens when O_2 concentration changes.

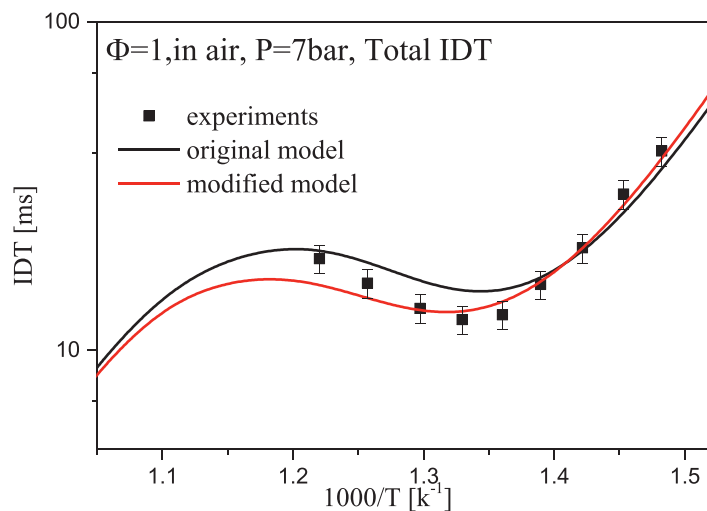
3.2.4. Correlation of total IDTs and first-stage IDTs

The IDTs correlation can be a quantitative method for describing the effect of operating conditions such as temperature (T_c), compressed pressure (P_c), fuel content (X_{fuel}) and oxygen content (X_{O_2}) on the ignition delay times (τ) of the reactant mixture. Both total IDTs in the low-T region and first-stage IDTs exhibit Arrhenius-like temperature dependence, and the correlation formula is expressed as follows:

$$\tau = A \cdot [P_c]^a [X_{fuel}]^b [X_{O_2}]^c \cdot \exp\left[\frac{E_a}{RT_c}\right] \quad (4)$$

Table 4
Modifications of several reactions in the mechanism.

NO	Reactions	Original Parameters			Modifications	Reference
		A	n	E _a [cal/mol]		
01	NC16-OQOOH=>CH2O+5.E-01NC5H10+1.05NC10H20+OH+CH3CO	1.85E+14	0	3.87E+04	E _a = 4.00E+04	[38,39]
02	NC16-OQOOH=>C2H5CHO+1.E-01NC5H10+2.E-01NC7H14+9.1E-01NC10H20+OH+CH3CO	1.85E+14	0	3.87E+04	E _a = 4.00E+04	[38,39]
03	NC16-OQOOH=>8.4E-01CO+CH3CHO+1.1NC10H20+OH+8.4E-01C2H5+1.6E-01CH3COCH2	5.00E+14	0	3.87E+04	E _a = 4.00E+04	[38,39]
04	NC16-OQOOH=>8.4E-01CO+C3H7CHO+9.E-01NC10H20+OH+8.4E-01C2H5+1.6E-01CH3COCH2	1.30E+14	0	3.87E+04	E _a = 4.00E+04	[38,39]
05	KHDECA=>CH2CO+CYC6H10+OH+CH2CHO	1.00E+15	0	4.3E+04	E _a = 4.00E+04	[38,39]
06	O2+NC16H33=>NC16H33-OO	2.00E+12	0	0	A*2	[40]
07	O2+RDECALIN=>RDECOO	1.00E+12	0	0	A*2	[40]

**Fig. 9.** Comparison of the IDTs from RCM and the simulation result using the original mechanism and the modified mechanism at the condition of $\Phi=1$, $P=7$ bar.

where a , b , c , and E_a are the indicators of the IDTs independence for their corresponding factors in Eq. (4), respectively. According to the definition, a higher absolute value of the above indicator means a stronger effect on the system reactivity.

The normalized ignition delay times ($\tau/([P_c]^a[X_{fuel}]^b[X_{O_2}]^c)$) and the correlation results are displayed in Fig. 7. The correlation coefficient R^2 of total IDTs and 1st IDTs are 0.97 and 0.98, respectively. Overall, the proposed correlation equation reproduces the measured IDTs satisfactorily, thus the correlation equation can be used to empirically estimate the IDTs within the range of tested conditions. Consistent with the experimental trend, the correlation formula also demonstrates that both total IDTs and 1st IDTs decrease with the increase of pressure, fuel content and oxygen content. Among the three parameters, pressure exhibits the biggest influence on ignition delays and fuel content shows the lowest influence. Moreover, 1st IDTs shows a stronger dependence on temperature compared to total IDTs. The results agree well with the observation in Fig. 4 to Fig. 6, in which the slopes of 1st IDTs against temperature all are greater than the corresponding slopes of total IDTs in the low-T region.

4. Model simulation and kinetic analysis

4.1. Kinetic mechanism

The kinetic mechanism used in this study was developed by the CRECK modeling group (<http://creckmodeling.chem.polimi.it/>) [37].

The kinetic models for the selected components were less developed since the four components are all large hydrocarbons [17,24–26]. To our knowledge, the CRECK model is the only mechanism that contains the low-temperature reactions of all the above components. The chosen model consists of 243 species and 6226 reactions, and the name, formula and brief description of all the species in the current model has been provided in the supplemental material, referenced from [37]. Besides, it is also worth mentioning that the CRECK model probably could be the most suitable model to extend to heavier compounds (C20) due to the lumping approach strongly reducing the number of species.

Based on the preliminary analysis, it is found that the predicting quality of the chosen model in the NTC range and the Low-T range still needs to be improved. So as to recognize the crucial reactions, sensitivity analysis using the original model is performed, and Fig. 8(a) and (b) show the calculation results at 680 K and 780 K, respectively. Then, several reactions (indicated with bolder fonts in Fig. 8) were modified to improve the predicting quality. The principles of adjustments are mainly twofold: first, the adjusted reaction may highly affect the overall ignition based on the sensitivity analysis, and second, the rate constants for these reactions were estimated by analogies and they are far from reliable. As was shown in Table 4, the modifications can be mainly divided into two parts. One of the changes to the mechanism was to adjust the activation energy of the ketohydroperoxide decomposition. As is demonstrated in Fig. 8(a), the decomposition of ketohydroperoxides (NC16-OQOOH and KHDECA) have an obvious in-

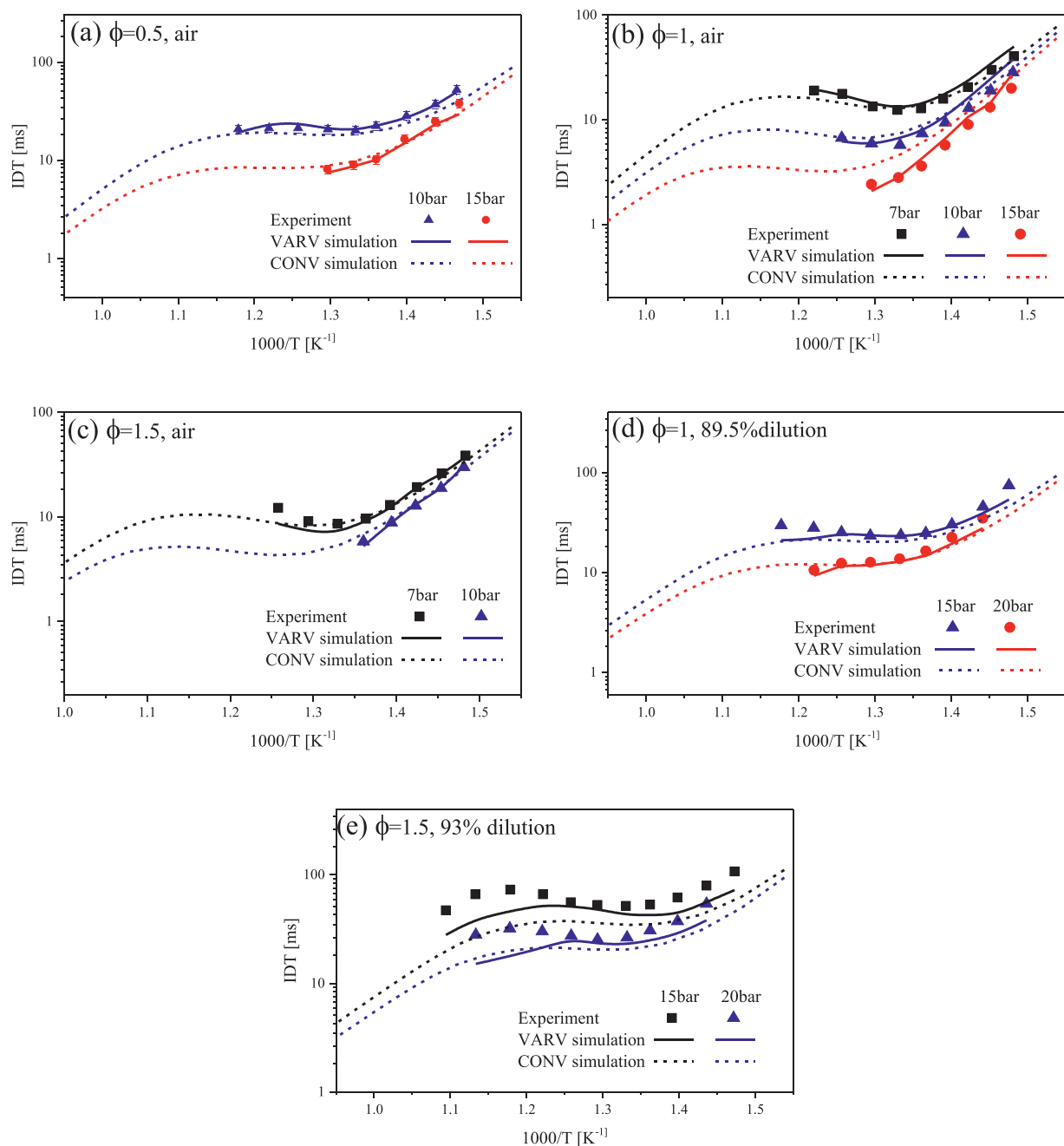


Fig. 10. Comparison of measured and simulated total ignition-delay-times of surrogate fuel M2.

fluence on the Low-T ignition process. According to the previous studies [38,39], the activation energy of this reaction class was assigned from 39 kcal/mol to 44.5 kcal/mol, and fundamental experiments and calculations are needed to obtain a more accurate estimate of this rate constant. In this study, the activation energy of five reactions (reaction 01–05 in Table 4) was set as 40 kcal/mol based on the IDT test results in the current study. Meanwhile, the addition of fuel radicals to molecular oxygen O_2 initiated the low-T chain branching reactions of hydrocarbon fuels, and thus has a significant influence on the whole ignition process. Therefore, this study modified the rate constants of the first O_2 addition to NC16H33 and RDECALIN (reaction 06 and 07 in the following Table). It should be pointed out that the modified parameters are all directly related to the reaction of macromolecular groups such as n-hexadecane and decalin. These parameters are mostly estimated based on the similar kinds of reaction constants from smaller

molecules, and the uncertainties of these parameters are rather large.

Fig. 9 shows the comparison of the RCM ignition delay time and the simulation result using the original mechanism and the modified mechanism at the equivalence ratio of 1 under the pressure of 7 bar. As it was illustrated in the figure that the predicting performance of the tuned mechanism was better than the original in the NTC range and the low-T range. The validations of models against the experimental data and the following kinetic analysis were carried out using the tuned mechanism.

4.2. Model validation

In Fig. 10 and Fig. 11, all the measured values for total IDTs (solid scattered points) and the 1st IDTs (hollow scattered point) are shown, with their corresponding simulation results. As was in-

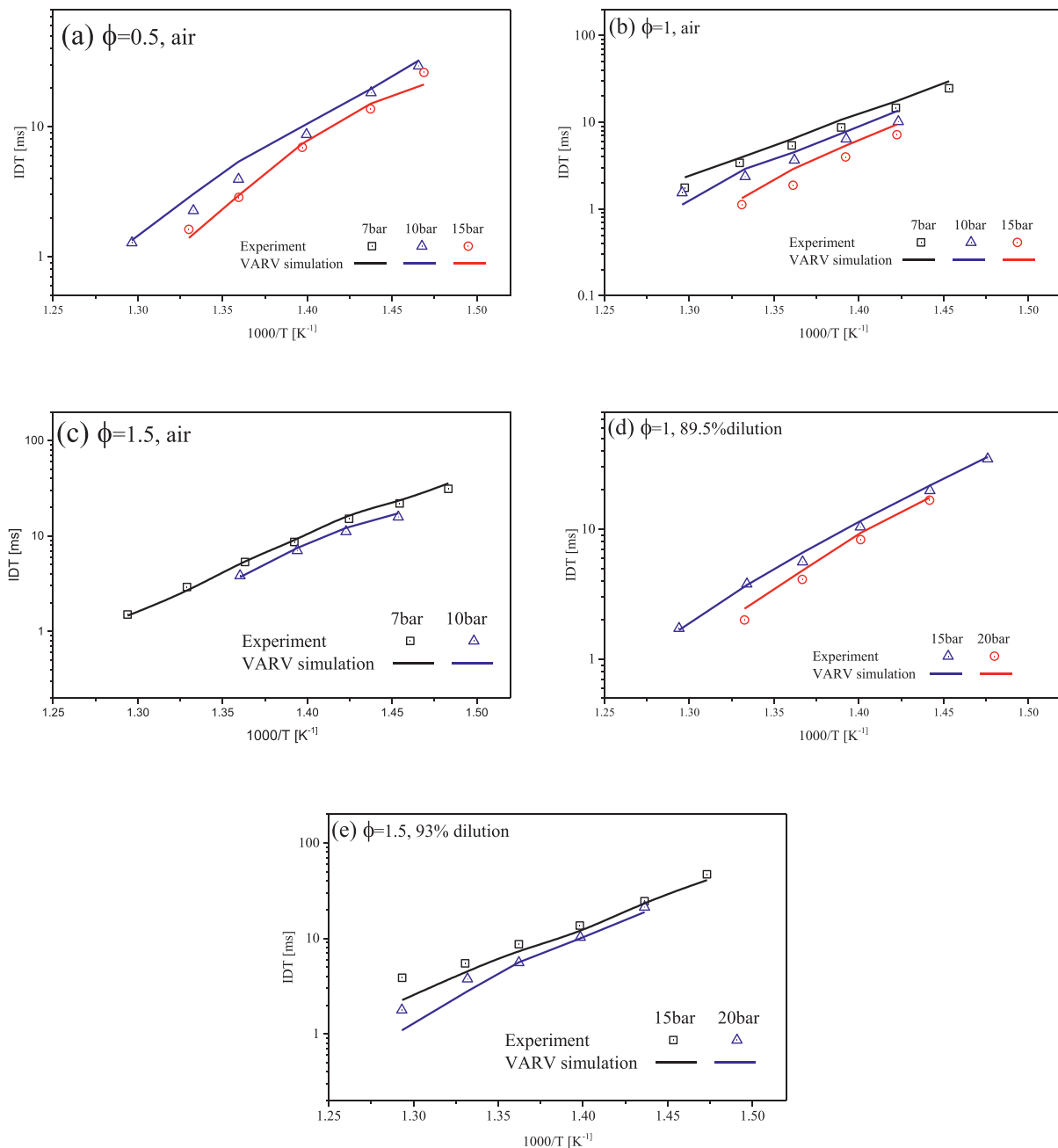


Fig. 11. Comparison of measured and simulated first-stage ignition-delay-times of surrogate fuel M2.

roduced in Section 2.3, constant volume simulation (CONV) and variable volume simulation (VARV) were performed to predict the auto-ignition characteristics of M2. The VARV simulation results (solid line) of total IDTs and 1st IDTs as well as the CONV simulation results (dashed line) of total IDTs are presented. On the whole, the optimized kinetic model can satisfactorily predict the IDTs in all the test conditions, and the model is able to capture the tendency of IDTs in the entire temperature range as well as the ignition data dependence on pressure and fuel/oxygen content. It is also noticed that the predicting quality of VARV results is better than the CONV results, which further proves that the validity of the modification since the simulation in VARV method is closer to the real process in RCM. Comparing the results from the two simulation methods, normally, the VARV results would be higher than the CONV results. This is because the heat loss in real RCM is counted into the VARV process, which will lower the reactivity of the sys-

tem and prolong the ignition delay times. However, in some cases, it is observed that the VARV IDT values are shorter than that of CONV at some temperature points. This may result from the promoting effect of the initial radicals produced in the compression process, and this enhancement is stronger than the inhabitation of counted heat loss, so the IDTs from VARV would be shorter than IDTs of CONV.

However, optimizations are still needed to improve the predicting ability of the current model. As shown in Fig. 10, though the model can reproduce the typical auto-ignition characteristics of M2, Slight discrepancies (mostly about 20%) still exist in the NTC-to-intermediate region between the simulations and the measured data. Meanwhile, the predictions of 1st IDTs of some test conditions also exhibit deviations with the experimental results. Thus, wider ranges of validations should be carried out to improve the fidelity of the mechanism in the future.

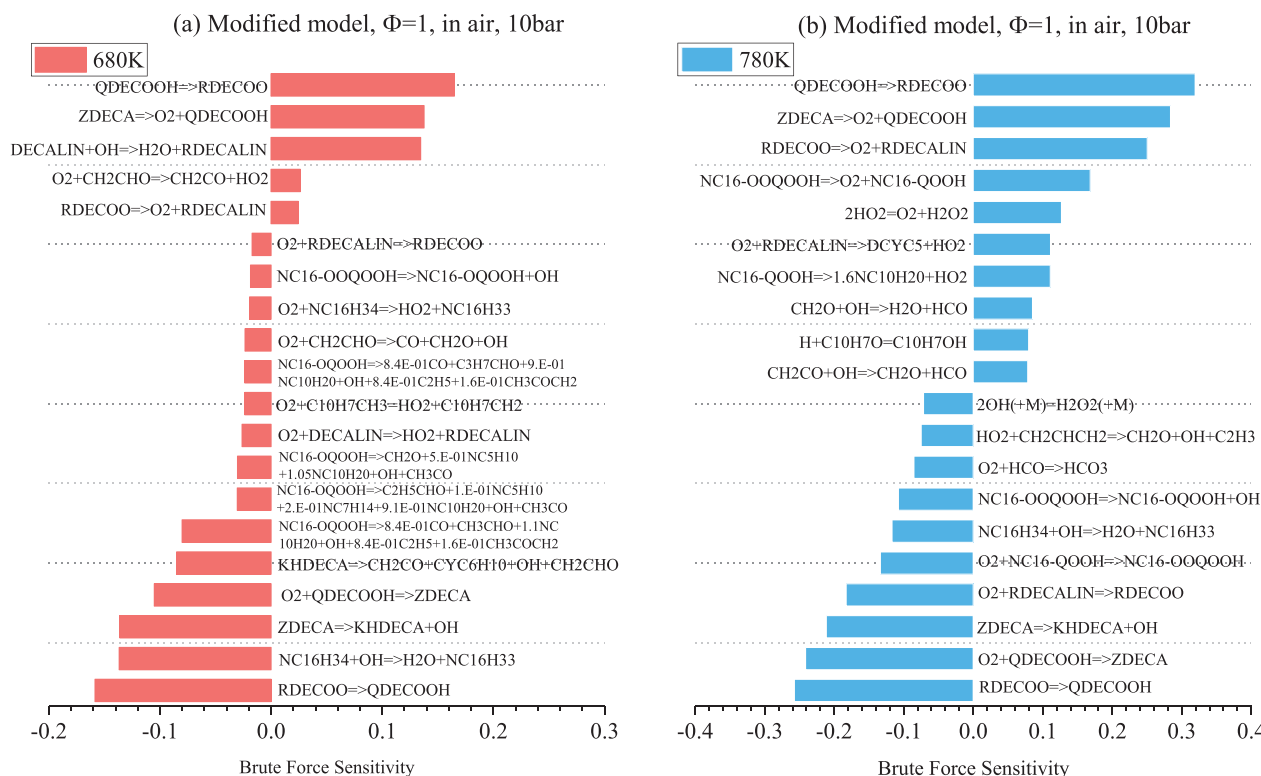


Fig. 12. Brute-force sensitivity analyses of surrogate M2 using the modified model: (a) 680 K, (b) 780 K.

4.3. Sensitivity analysis

To identify the dominant reactions in the auto-ignition process, brute force sensitivity analysis of total IDTs was performed. Two temperatures, 680 K and 780 K, were selected to represent low-T region and NTC region, respectively. The analysis results are displayed in Fig. 12. Generally, the elementary reactions associated with n-hexadecane and decalin contained in M2 have a great influence on the ignition delays. This may due to their relatively higher proportion in M2 and the higher CN of these two components. Both at the temperature of 680 K and 780 K, the low-T chain-branching reactions related to decalin, including: $\text{RDECOO}/\text{QDECOOH}$ isomerization, ZDECA (a hydroperoxyalkyl peroxide radical) decomposition and the O_2 -addition to QDECOOH radical, exhibit strong negative sensitivities, which means increasing the rate constants of these reactions could significantly reduce the ignition delay times. At the same time, the reverse reactions of the above three reactions all show a large positive sensitivity at the two temperatures. Hence, the low-T chain-branching pathways of decalin greatly affect the overall reactivity of the surrogate fuel. Meanwhile, the H-atom abstraction of n-hexadecane by OH radicals $\text{NC16H34} + \text{OH} \Rightarrow \text{H}_2\text{O} + \text{NC16H33}$ also has a large negative sensitivity at both temperatures. This may because the produced Hexadecyl (NC16H33) initiate the low-temperature chain-branching pathways of n-hexadecane. Besides, reactions, including the O_2 -addition to NC16-QOOH and the isomerization reaction ($\text{NC16-QOOH} \Rightarrow \text{NC16-QOOH} + \text{OH}$) of n-hexadecane, as well as the O_2 -addition to the alkyl radical (RDECALIN) and small species like HCO, have a significant promotion effect on the reaction activity at 780 K, but the effect becomes weaker at 680 K. On the contrary, the dissociation of ketohydroperoxides (the modified reactions displayed in table 4) possess a higher negative sensitivity at 680 K than that at 780 K, indicating that these reactions have higher influence in Low-T range than in NTC region.

As for the reactions with positive sensitivities, in addition to the reverse reaction of several low-T reactions of decalin, the H-abstraction of decalin by OH ($\text{DECALIN} + \text{OH} \Rightarrow \text{H}_2\text{O} + \text{RDECALIN}$) at 680 K also presents an significant inhibitory effect on the overall reactivity. While for the condition under the temperature of 780 K, the reverse reaction of the second-addition of O_2 to NC16-QOOH ($\text{NC16-QOOH} \Rightarrow \text{O}_2 + \text{NC16-QOOH}$) and the decomposition of NC16-QOOH ($\text{NC16-QOOH} \Rightarrow 1.6\text{NC10H20} + \text{HO}_2$) as well as the H-abstraction reaction of RDECALIN ($\text{O}_2 + \text{RDECALIN} \Rightarrow \text{DCYC5} + \text{HO}_2$) also significantly inhibit the system reaction activity. At the same time, the chain termination reaction $2\text{HO}_2 = \text{O}_2 + \text{H}_2\text{O}_2$ and the small molecule reactions consuming OH radicals $\text{CH}_2\text{CO} + \text{OH} \Rightarrow \text{CH}_2\text{O} + \text{HCO}$ and $\text{CH}_2\text{O} + \text{OH} \Rightarrow \text{H}_2\text{O} + \text{HCO}$ also have large positive sensitivity and would greatly inhibit the system activity. Also, it should be pointed out that the inhibitory effect of the above listed reactions on the system activity at 780 K is significantly stronger than that at 680 K, which means that the same variation in fuel or oxygen concentration will lead to a larger IDT difference in the NTC region. This finding may could explain the enhanced IDT dependence on pressure and reactant concentration in the NTC range, as illustrated in Section 3.2.

5. Conclusion

The present study provides the IDTs of a large four-component surrogate fuel over a wide range of conditions (pressure: 7, 10, 15, 20 bar; equivalent ratio: 0.5, 1, 1.5; temperature: around 675–914 K). The surrogate fuel M2, containing 17.3% n-hexadecane, 5.93% 2,2,4,4,6,8,8-heptamethylnonane, 30% decalin and 46.77% 1-methylnaphthalene by mole fraction, is developed for the near-term modeling of the heavy marine diesel. The experiments were conducted on a rapid compression machine in SJTU. Test results showed obvious two-stage ignition phenomena and NTC behaviours under different operating conditions. The effects of

compressed pressure, equivalence ratio and oxygen content were systematically investigated and quantitatively explained with correlation formulas. Both total IDTs and 1st IDTs are decreasing with the rise of the above three parameters, and the influence of the above parameters on IDTs all get higher as the temperature moves to the intermediate-T range from low-T range. Meanwhile, the correlation demonstrates that pressure shows the highest influence on the reactivity of the system compared to another two factors, and total IDTs shows a stronger dependence on temperature than 1st IDTs in the low-T region.

In order to improve the predicting quality of an existing model for surrogate M2, several modifications were made in some reactions according to two principles: (a) the adjusted reactions have large sensitivities based on the sensitivity analysis; (b) the rate constants were estimated by analogies and had larger uncertainties. Constant volume simulation (CONV) and variable volume simulation (VARV) were performed to predict the auto-ignition characteristics using the modified model, and the simulation results were validated against the measured IDTs to evaluate the performance of the model. The results exhibited that the modified model could satisfactorily reproduce the total IDTs and the 1st IDTs under the test conditions, and meanwhile, the model could also well describe IDT dependence on the operating conditions. Finally, key reactions controlling autoignition and potential reactions for further model improvement were identified through brute force sensitivity analyses at low-temperature (680 K) and NTC temperature (780 K). Overall, this paper provides very plentiful and precious IDT data of the large four-component surrogate fuel for model validation as well as further CFD computation for the marine diesel engine.

Declaration of Competing Interest

None.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2021.02.003](https://doi.org/10.1016/j.combustflame.2021.02.003).

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